NEURO 208, School of Medicine, University of Crete

Lecture 2: Spiking neuronal models

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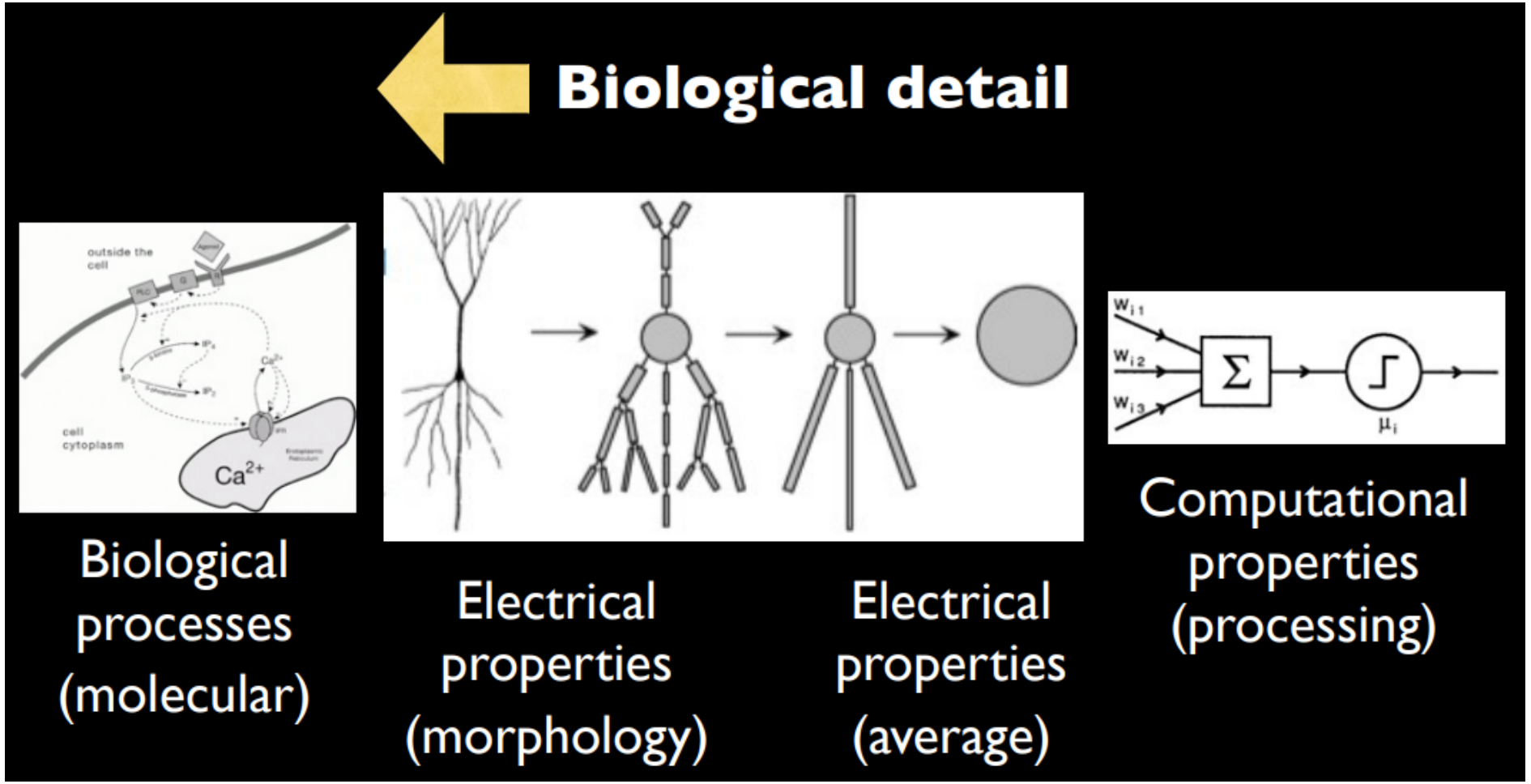
IMBB-FORTH

Appropriate level of models

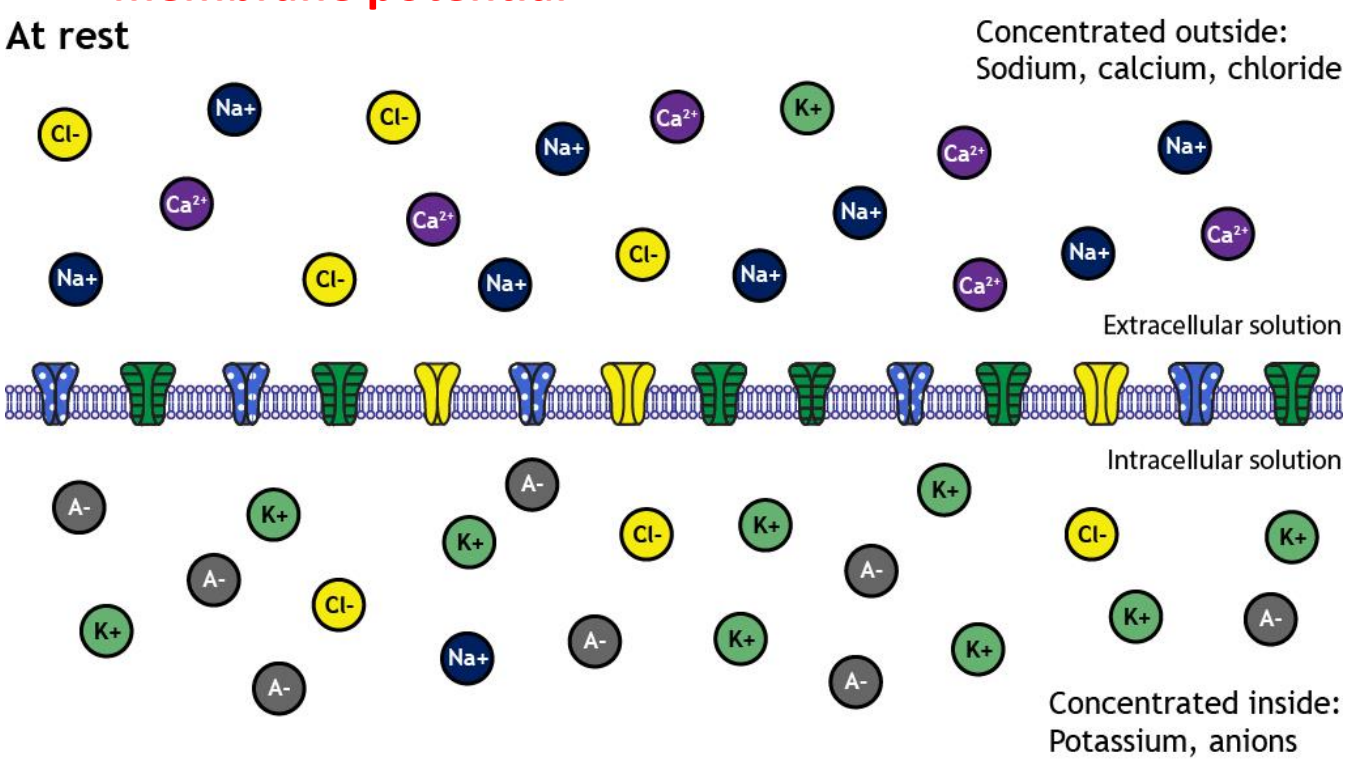


Classification of neuron models

- Point neuron models (no geometry)
  - Spiking neurons
    - McCulloch-Pitts
    - Integrate and Fire
    - Hodgkin-Huxley
  - Rate neurons
    - McCulloch-Pitts
    - Artificial Neural Networks
    - Single Cells
- Multi-compartmental models (geometry)

Ions and the membrane

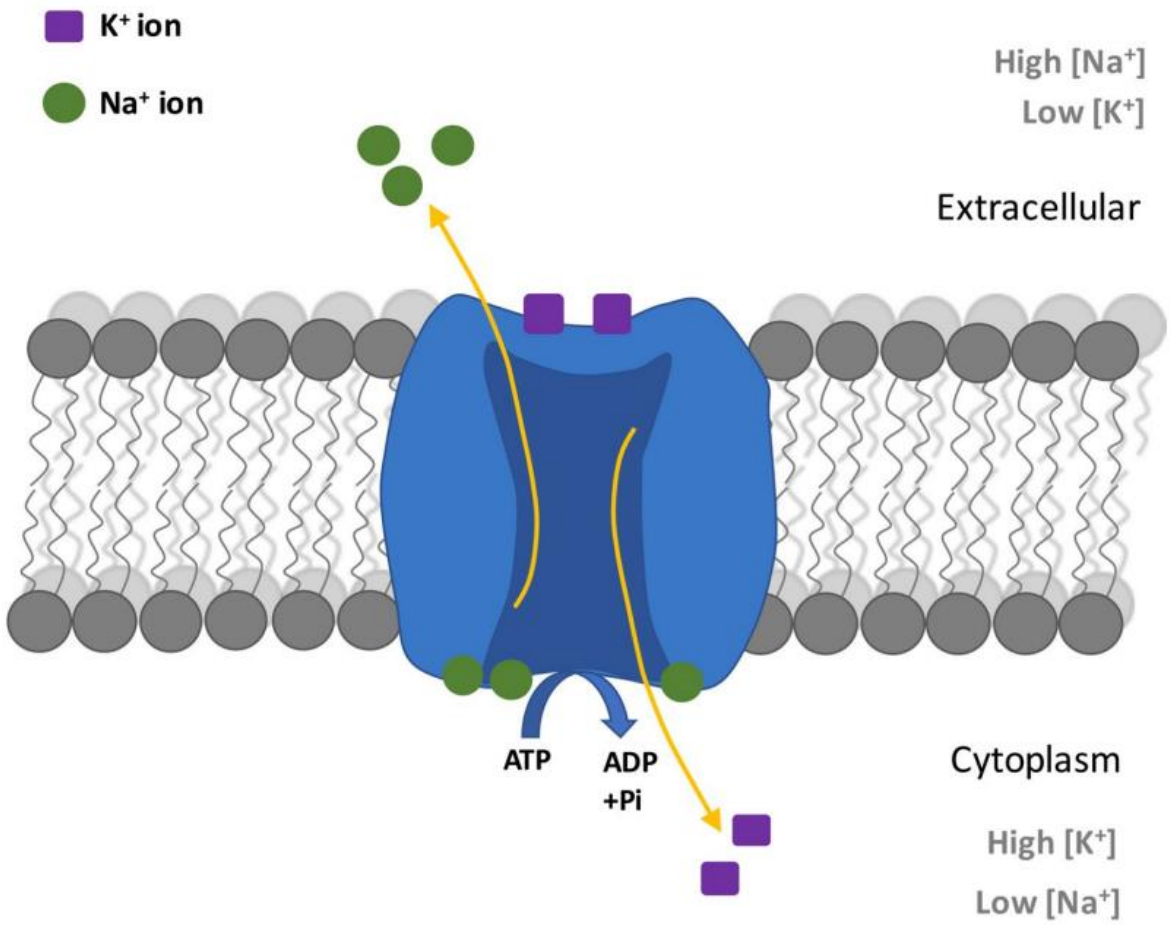
- Neurons are enclosed by a membrane separating interior from extracellular space
- The concentration of ions of Na^+, K^+, Ca^{2+}, Cl^- inside is different to that in the surrounding liquid (thanks to the "$3Na^+$ out & $2K^+$ in" pump)
- The difference in ionic concentrations (*in* vs. *out*) generates an electrical potential (voltage), called **membrane potential**



Source: https://openbooks.lib.msu.edu/neuroscience/chapter/the-membrane-at-rest/

$$V_{\mathrm{m}} = V_{\mathrm{in}} - V_{\mathrm{out}} \approx -65\mathrm{mV}$$

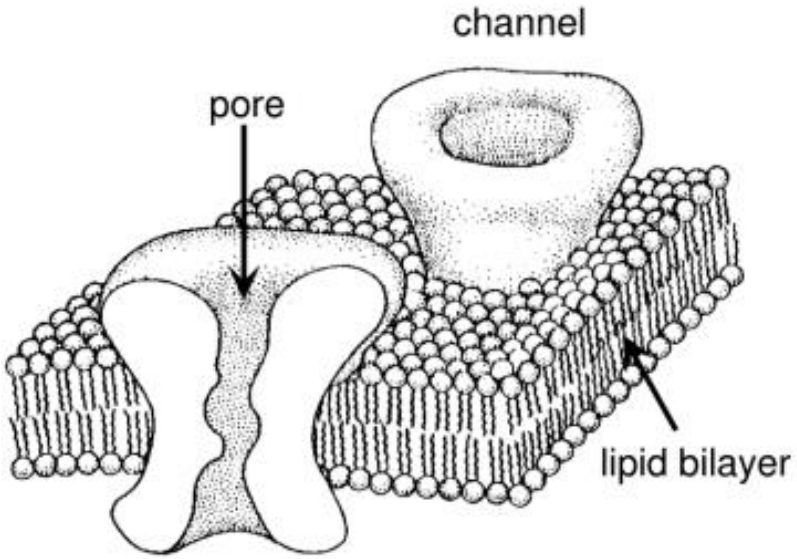
Na/K pump



Source: [Pivovarov et al., 2018]

Types of ion channels in the neuron membrane

- **Permanently open or "passive" ion channels**: are everywhere. They allow "passive" flow of ions along the electro-chemical gradients. Passive ion channels underlie the theory of dendrites
- **Voltage-gated ion channels**: are located in the soma and axon, and play role in generation and propagation of action potentials. They open and close due to the changes in the membrane potential V_m
- **Receptor-gated ion channels**: are located on the dendrites and soma. They open only when a particular chemical binds to them



Source: Theoretical Neuroscience

Equilibrium potential: Nernst equation

- Difference in concentration between inside and outside mean ions move through passive channels along their electro-chemical gradient
- Equilibrium potential E_X for an ion is the membrane potential at which current flow due to electric forces cancels diffusive flow
- Nernst Equation:

$$E_X = \frac{RT}{z_X F} \ln\left(\frac{[X]_{\text{in}}}{[X]_{\text{out}}}\right)$$

Where z_X is the electrical charge (or valence) of the ion X, R is the universal gas constant ($8.314\ J\ K^{-1}\ \text{mol}^{-1}$), T is the absolute temperature (K) and F is the Faraday constant ($9.648 \times 10^4\ C\ \text{mol}^{-1}$). $[\cdot]$ denotes the concentration of the ion X in and out of the membrane, respectively.

Equilibrium potential: Nernst equation

$$E_X = \frac{RT}{z_X F} \ln\left(\frac{[X]_{in}}{[X]_{out}}\right)$$

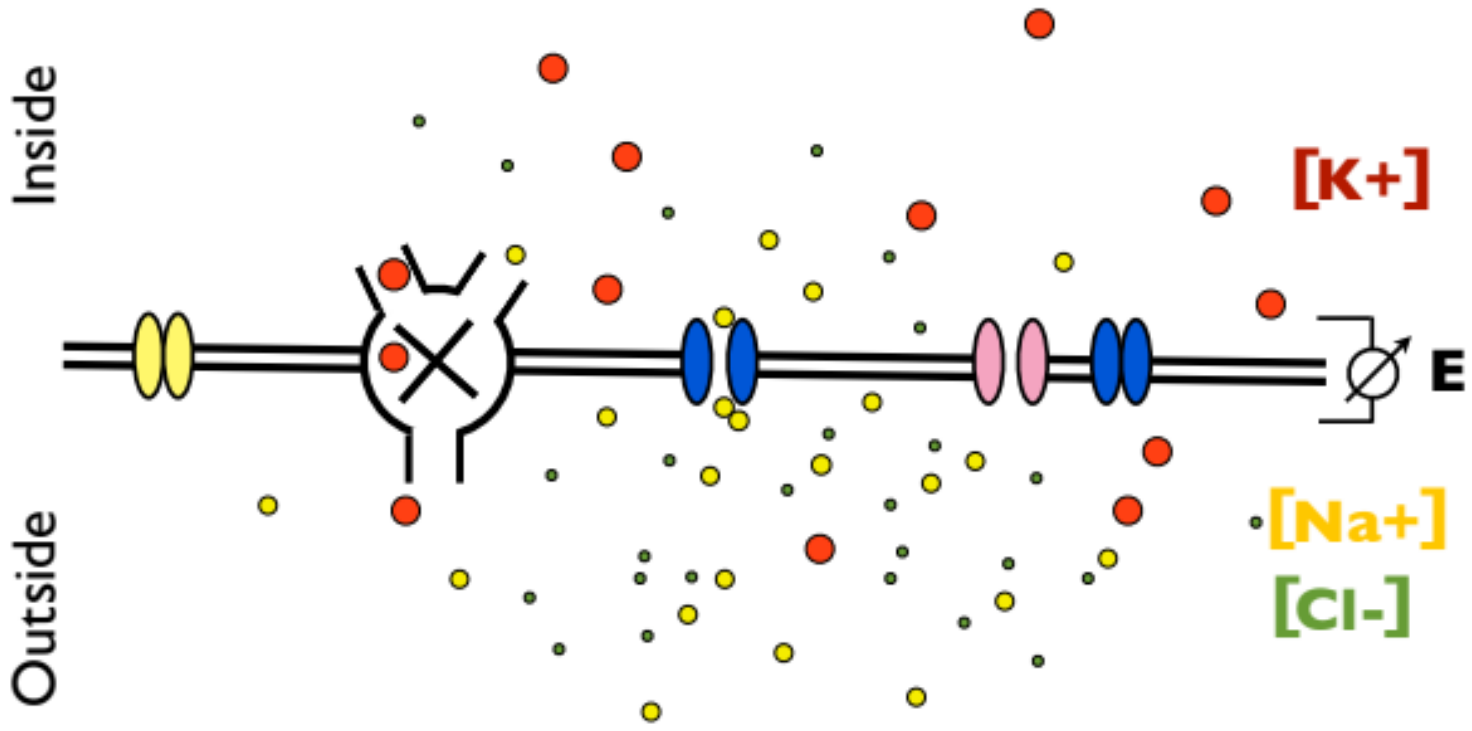
- From the Nernst equation we get these equilibrium potentials:
  - $\mathrm{E_K}$ is typically between –70 and –90 mV
  - $\mathrm{E_{Na}}$ is +50 mV or higher
  - $\mathrm{E_{Ca}}$ is around +150mV
  - $\mathrm{E_{Cl}}$ is about – 65mV (near the resting potential of many neurons)
- If $V > E_K$ then K^+ ions flow out from neuron and **hyperpolarize** it
- if $V < E_{\mathrm{Na}}$ or E_{Ca}, then these ions flow in and **depolarize** neuron

Resting potential: GHK equation

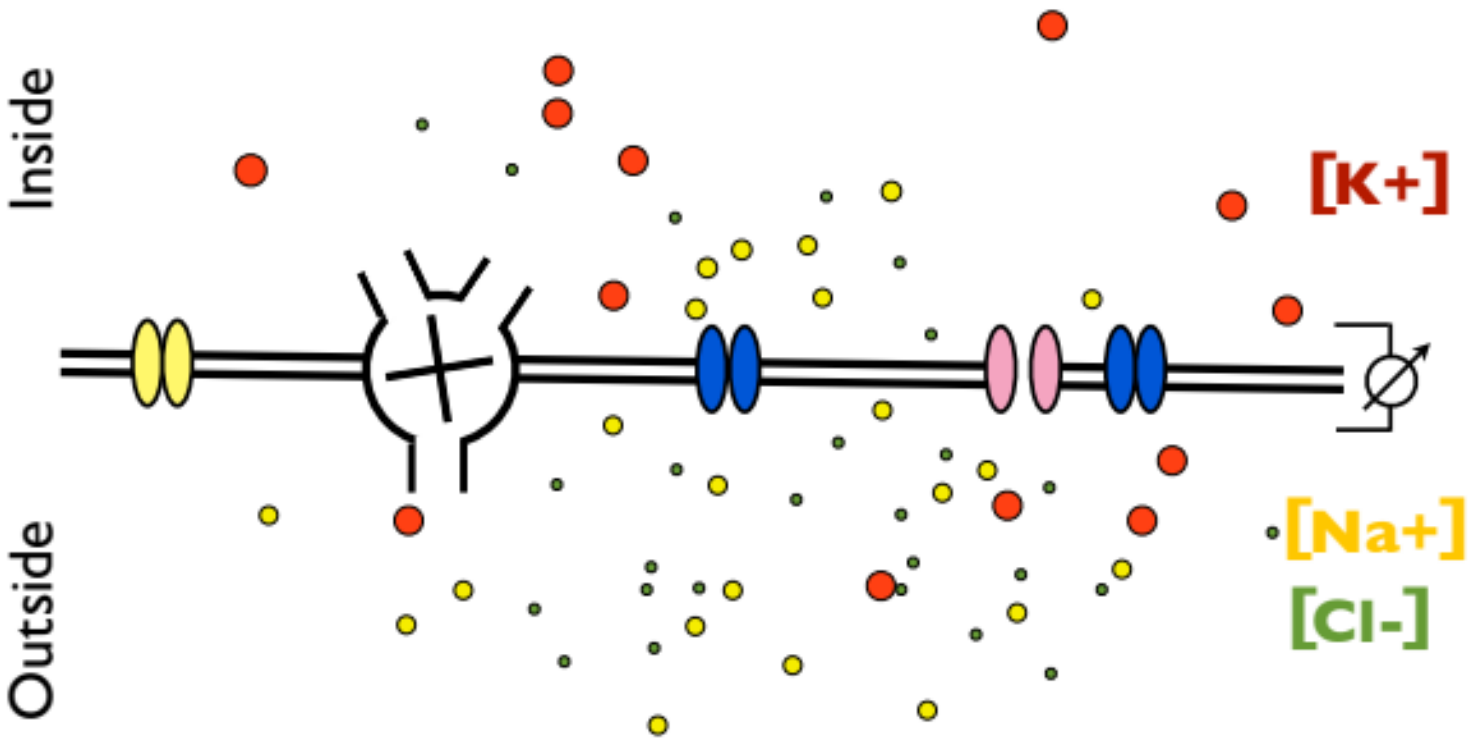
- When the membrane is in thermodynamic equilibrium (i.e., no ion flux), the membrane potential is equal to the Nernst potential E.
- However, in physiology, due to active ion pumps, the inside and outside of a cell are not in equilibrium and there are many ions with different E.
- thus the resting potential V_{rest} is determined from the **Goldman-Hodgkin-Katz (GHK)** equation (P_X is the permeability for ion X):

$$V_{rest} = \frac{RT}{F} \ln \left(\frac{\sum_{X^+} P_{X^+} [X^+]_{\text{out}} + \sum_{X^-} P_{X^-} [X^-]_{\text{in}}}{\sum_{X^+} P_{X^+} [X^+]_{\text{in}} + \sum_{X^-} P_{X^-} [X^-]_{\text{out}}} \right)$$

Recap



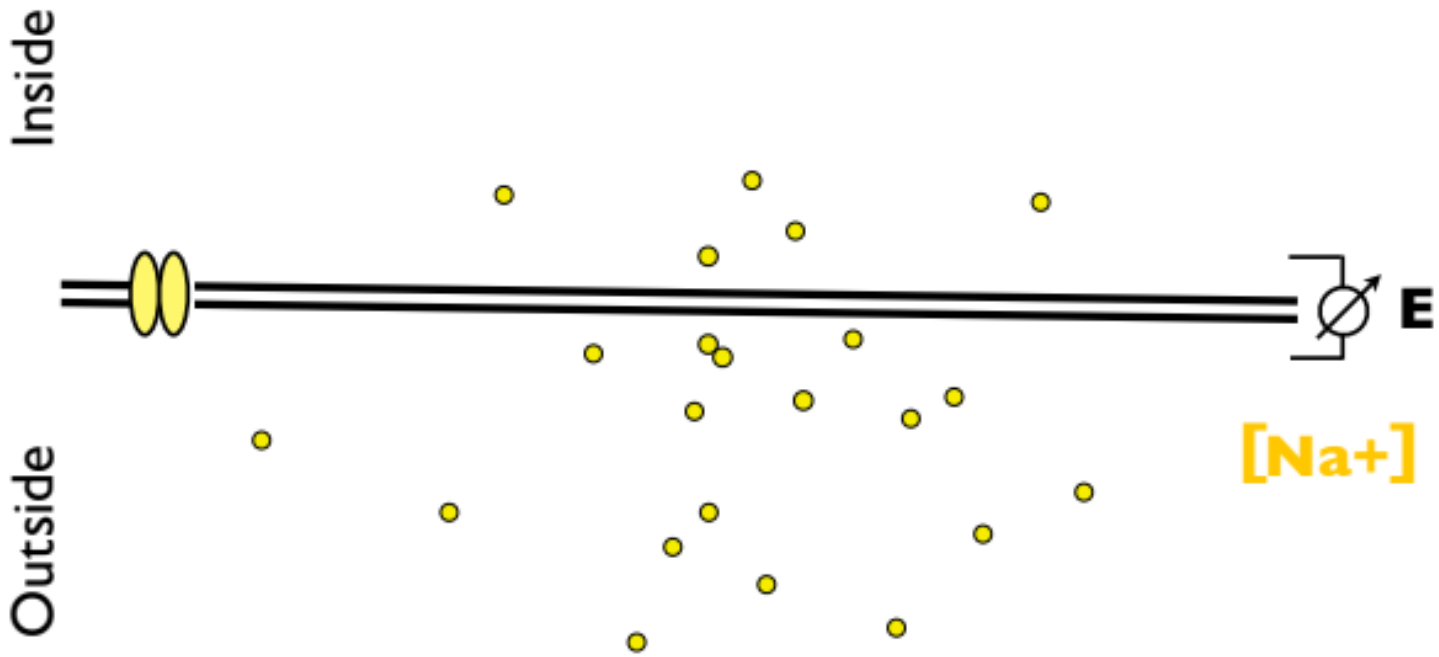
Recap



Recap

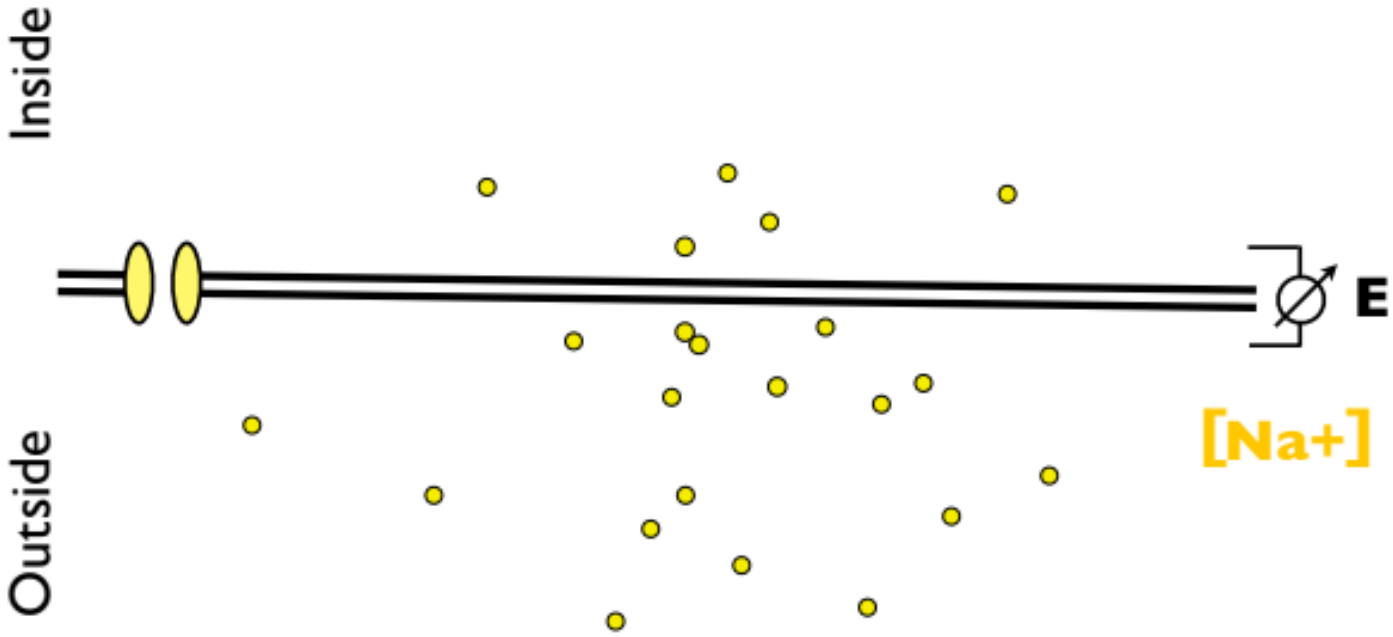
$$\mathbf{E} = K \ln\left(\frac{[Na+]_{out}}{[Na+]_{in}}\right)$$

Nernst equation, 1887



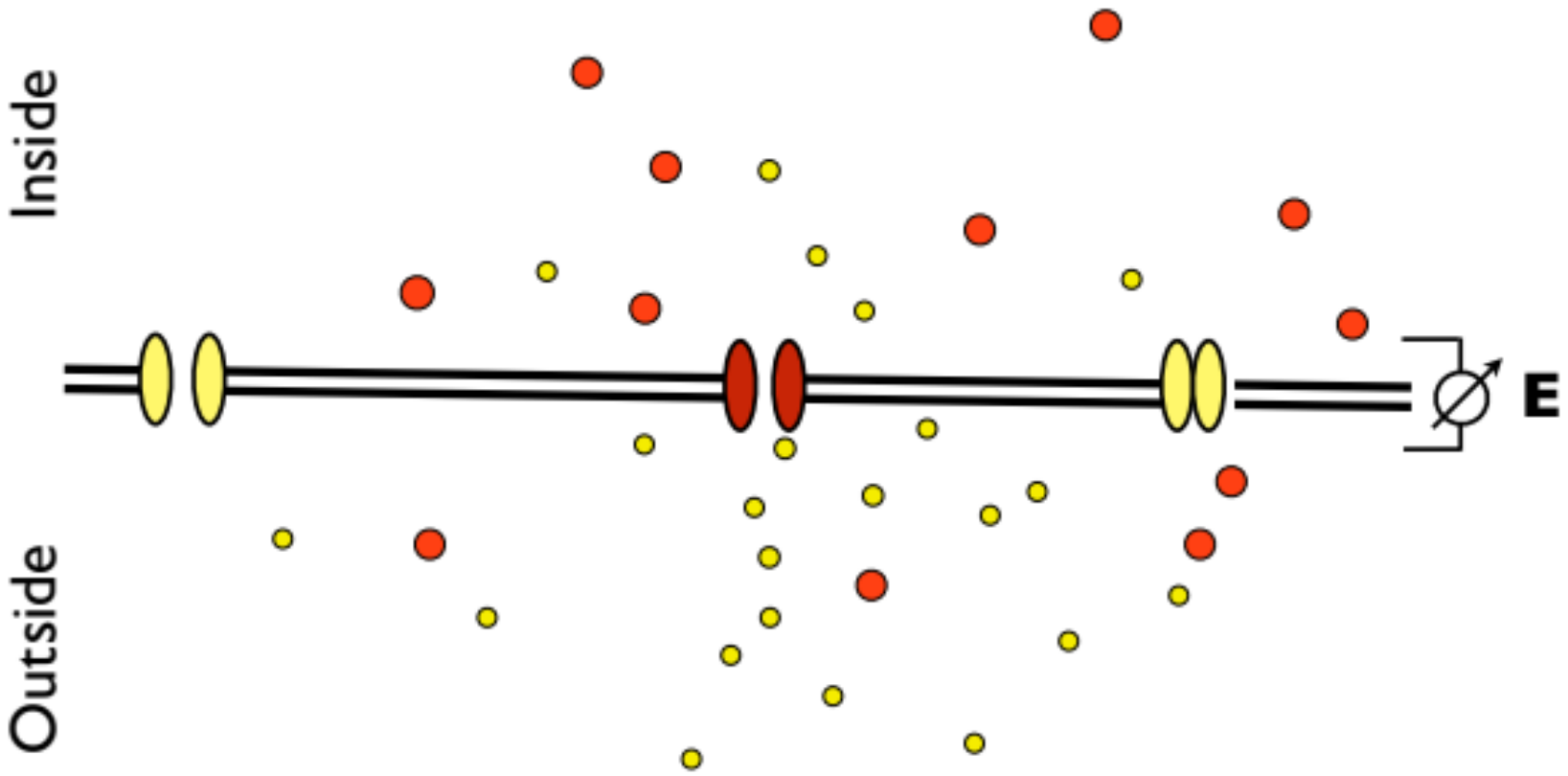
Recap

$$\mathbf{E} = C \ln\left(\frac{P[Na+]_{out}}{P[Na+]_{in}}\right)$$



Recap

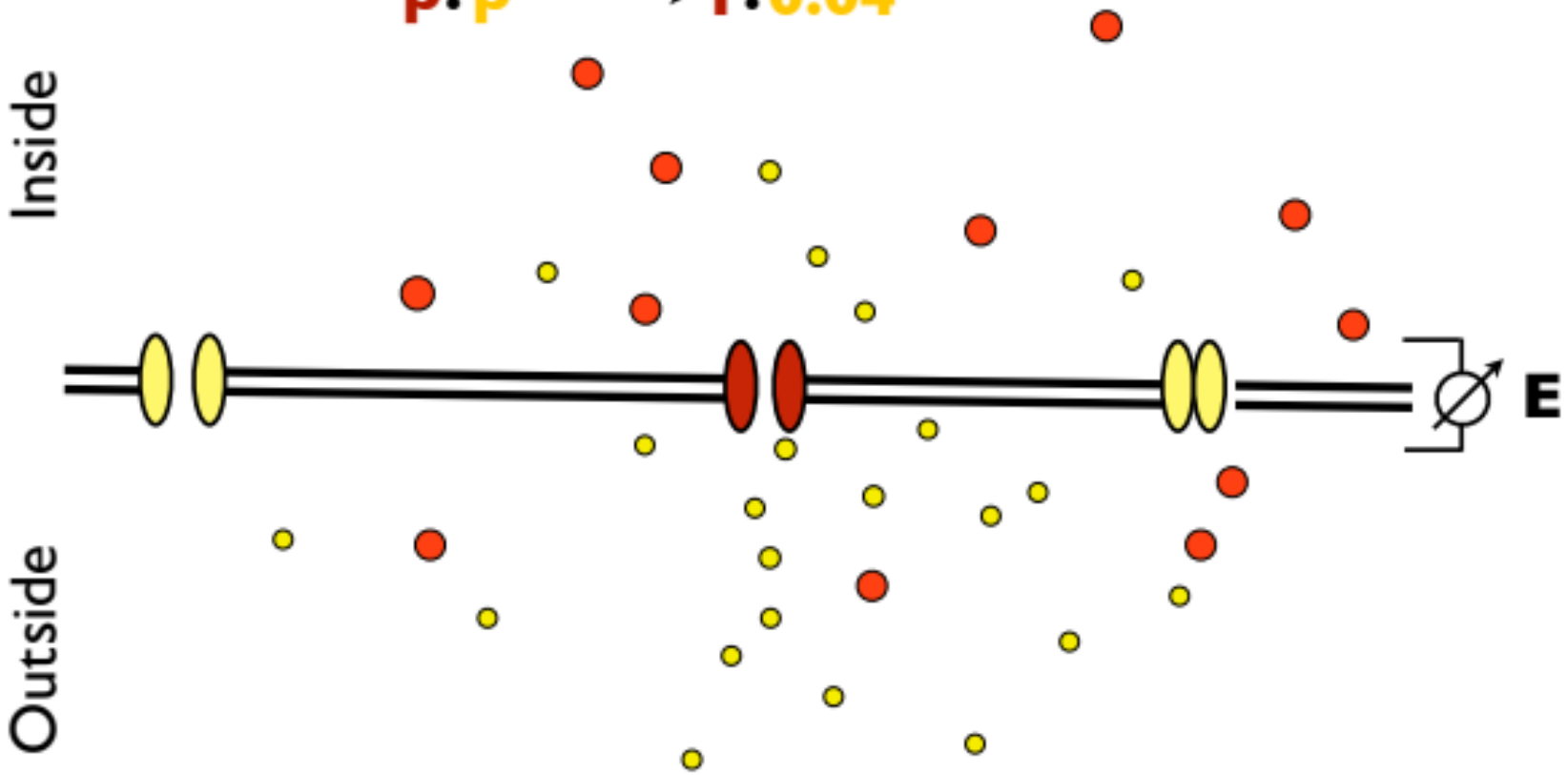
$$\mathbf{E} = C \ln\left(\frac{P[K+]_{out} + P[Na+]_{out}}{P[K+]_{in} + P[Na+]_{in}}\right)$$



Recap

$$E = C \ln\left(\frac{P[K+]_{out} + P[Na+]_{out}}{P[K+]_{in} + P[Na+]_{in}}\right)$$

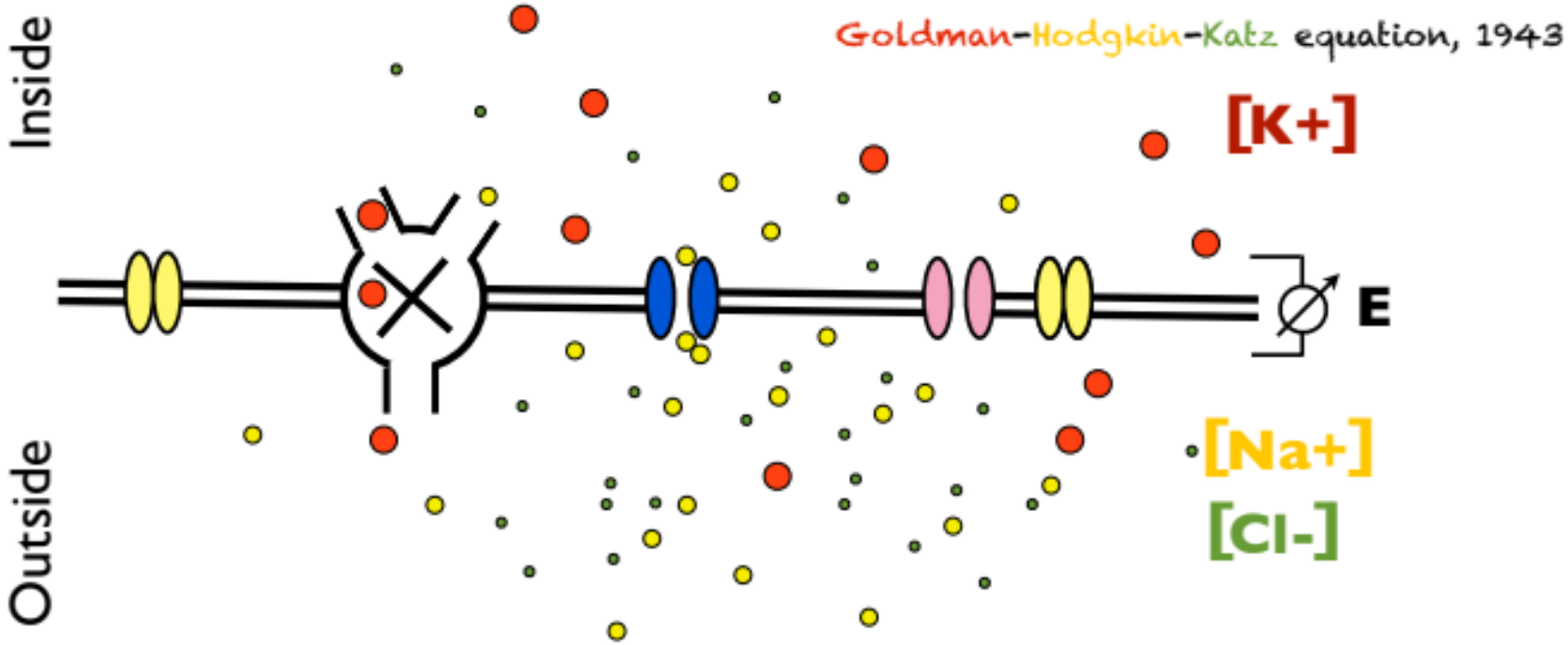
p:p ⟶ 1:0.04



Recap

$$E = C \ln\left(\frac{p[K+]_{out} + p[Na+]_{out} + p[Cl-]_{in}}{p[K+]_{in} + p[Na+]_{in} + p[Cl-]_{out}}\right)$$

$$p : p : p \longrightarrow 1 : 0.04 : 0.03$$



Resistance and capacitance of the membrane

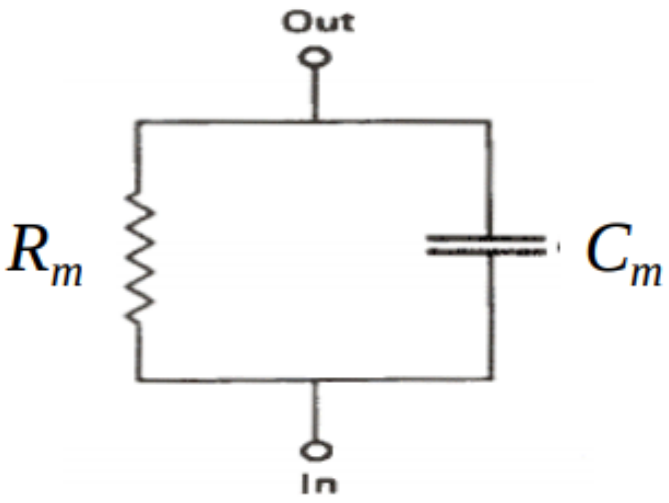
- Cell membrane: 2-3 nm thick and is impermeable to most charged molecules, thus acts as a **resistor** (however, the passive ion channels in the membrane lower the effective membrane resistance by a factor of 10,000)

- Membrane also acts as a **capacitor** by separating the charges lying on either side of the membrane

- Capacitors store charge across an insulating medium. They do not allow current to flow across, but charge can be redistributed on each side leading to the current flow

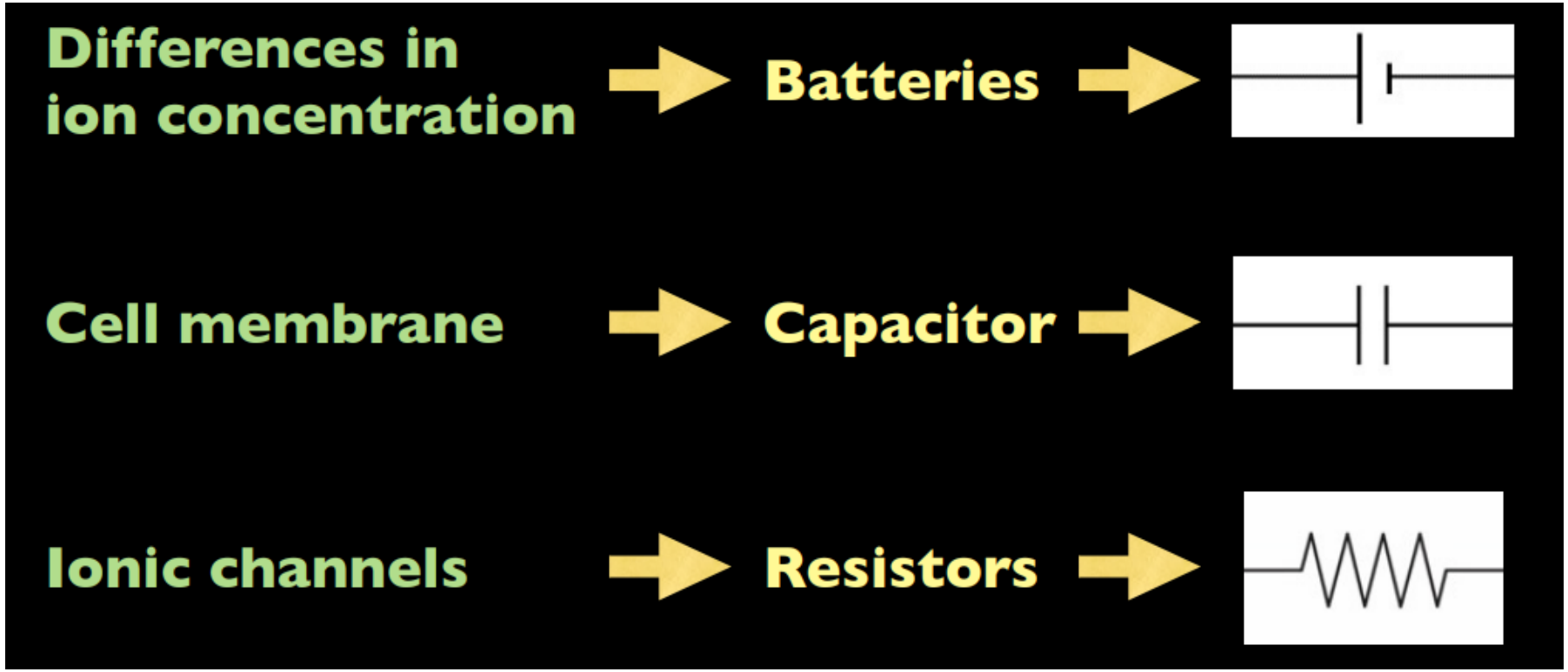
Electric circuit of the membrane patch

- The electric circuit representing a patch of the neuronal membrane looks like this:



- This is called an **RC circuit** in electronics since it consist of a resistor and capacitor
- What is a patch of membrane? Where R_{m} and C_{m} stay the same across time

Neuron as an electrical circuit



(Trans)membrane electric current I_m

- Let us denote the membrane capacitance as C_m and the electric charge on the membrane as Q. From physics we have: $Q = C_\mathrm{m} V$
- The electric current I_m through the membrane is equivalent to the change of electrical charge on the capacitor, i.e.:

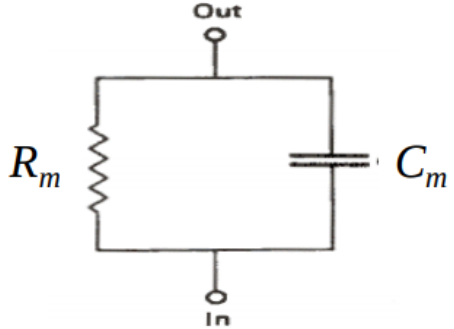
$$I_\mathrm{m} = \frac{\mathrm{d}Q}{\mathrm{d}t} = C_\mathrm{m} \frac{\mathrm{d}V}{\mathrm{d}t}$$

- Membrane also has a resistance R_m. Thus the membrane current I_m also equals to:

$$I_\mathrm{m} = \frac{V}{R_\mathrm{m}}$$

Electric circuit of the membrane patch

- In the RC circuit



- By convention capacitance current is ‘plus’ while membrane current is ‘minus’:

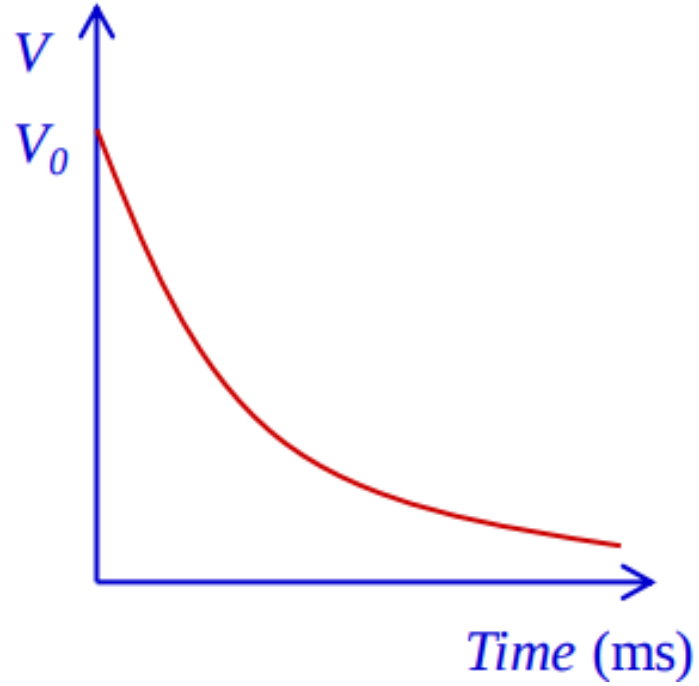
$$I_{\mathrm{m}} = C_{\mathrm{m}} \frac{\mathrm{d}V}{\mathrm{d}t} = -\frac{V}{R_{\mathrm{m}}}$$

- From which we derive an ODE for the voltage:

$$\frac{dV}{dt} = -\frac{V}{R_{\mathrm{m}} C_{\mathrm{m}}}$$

Elementary solution for the membrane voltage

- Solution of ODE for voltage: $$\frac{dV}{dt} = -\frac{1}{\tau_{\mathrm{m}}} V$$
- Is the exponential function: $$V = V_0 \exp\left(-\frac{t}{\tau_{\mathrm{m}}}\right)$$
- Where $V_0 = V(t = 0)$
- Thus if the initial voltage $V_0 > 0$, then V decays in time exponiall with the so-called membrane time constant $\tau_{\mathrm{m}} = C_{\mathrm{m}} R_{\mathrm{m}}$



McCulloch-Pitts

Integrate-and-Fire

Izhikevich model

Perfect Integrate-and-Fire (I&F) model

- One of the earliest models (Louis Lapicque, 1907)
- A neuron is represented by its membrane voltage V which evolves in time

$$C_{\mathrm{m}} \frac{\mathrm{d}V(t)}{\mathrm{d}t} = I_e(t)$$

where I_e is the stimulation current

- When an input current is applied, the membrane voltage increases with time until it reaches a constant threshold V_{th}

$$if\ at\ t, V \geq V_{th}\colon\ t_{spike} \rightarrow t, V \rightarrow V_{reset}$$

- The firing frequency of the model thus increases linearly without bound as input current I_e increases.

Leaky Integrate-and-Fire (I&F) model

- Contains a “leak” term in the membrane potential equation, reflecting the diffusion of ions through the membrane.

$$C_{\mathrm{m}} \frac{\mathrm{d}V(t)}{\mathrm{d}t} = I_e(t) - \frac{V(t) - E_L}{R_{\mathrm{m}}} = I_e(t) - \mathrm{g}_L(V(t) - E_L)$$

*When $R_{\mathrm{m}} \to \infty$ ($\mathrm{g}_L \to 0$), if the membrane is a perfect insulator, so we obtain the perfect I&F

- When $I_e = 0$, the membrane voltage returns exponentially to the resting potential E_L

- Based on the Euler formula:

$$V[t + dt] = V[t] + dt \cdot \left(\frac{I_e[t] - \frac{V[t] - E_L}{R_{\mathrm{m}}}}{C_{\mathrm{m}}} \right)$$

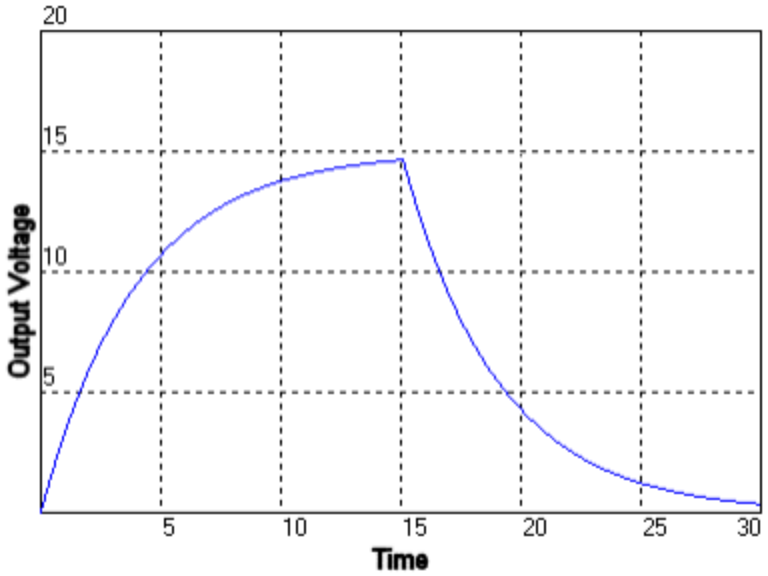
$$if\ V[t] \geq V_{th}: t_{spike} \to t, V[t] \to V_{reset}$$

Leaky Integrate-and-Fire (I&F) model

- Solution of the cable equation when we apply a rectangular current pulse

$$V = E_L + R_\mathrm{m} I_e \left(1 - \exp\left(-\frac{t}{\tau_\mathrm{m}}\right)\right)$$

- Looks like this:



Quadratic Integrate-and-Fire (I&F) model

- Integrate-and-fire is fast, but has a poor fit to the ionic current when V_{m} is close to the threshold
- Quadratic I&F: hybrid model, which models the ionic current close to the threshold better (Latham et al., 2000, Hansel and Mato, 2001, Zheng and Tonnelier, 2008)

$$C_{\mathrm{m}} \frac{\mathrm{d}V(t)}{\mathrm{d}t} = -\frac{(V(t) - E_{\mathrm{L}})\big(V_{\mathrm{th}} - V(t)\big)}{R_{\mathrm{m}}(V_{\mathrm{th}} - E_{\mathrm{L}})} + I_e(t)$$

$$if\ V[t] \geq V_{th}: t_{spike} \rightarrow t, V[t] \rightarrow V_{reset}$$

Exponential Integrate-and-Fire (I&F) model

- Non-linear variation of I&F: exponential I&F (Fourcaud-Trocmé et al, 2003)

$$C_{\mathrm{m}} \frac{\mathrm{d}V}{\mathrm{d}t} = -\left(\frac{V(t) - E_{\mathrm{L}}}{R_{\mathrm{m}}} - \frac{\Delta_{\mathrm{T}}}{R_{\mathrm{m}}} \exp\left(\frac{V(t) - V_{\mathrm{T}}}{\Delta_{\mathrm{T}}} \right) \right) + I_e(t)$$

V_{T}: threshold voltage

Δ_{T}: spike slope factor (determines the sharpness of initiation)

Adding adaptation

- In many biological neurons, the firing rate is response to a sustained current injection decreases throughout the spike train.
- Whenever the neuron spikes, the g_{adapt} is incremented by an amount $\Delta\mathrm{g}_{\mathrm{adapt}}$ and decays with a time constant τ_{adapt}

$$\frac{\mathrm{dg}_{\mathrm{adapt}}}{\mathrm{d}t} = -\frac{\mathrm{g}_{\mathrm{adapt}}}{\tau_{\mathrm{adapt}}}$$

$$I_{\mathrm{adapt}} = \mathrm{g}_{\mathrm{adapt}}(V - E_{\mathrm{L}})$$

$$At\ V(t) \geq V_{th}: t_{spike} \rightarrow t, V(t) \rightarrow V_{reset}, \mathrm{g}_{adapt} \rightarrow \mathrm{g}_{adapt} + \Delta\mathrm{g}_{adapt}$$

AdEx I&F: Adaptive Exponential I&F model

- Two-dimensional integrate-and-fire model that combines an exponential spike mechanism with an adaptation equation (Brette and Gerstner, 2005)

$$C_{\mathrm{m}} \frac{\mathrm{d}V(t)}{\mathrm{d}t} = -\mathrm{g}_L(V(t) - E_L) + \mathrm{g}_L \Delta_T \exp\left(\frac{V(t) - V_T}{\Delta_T}\right) - w(t) + I_e(t)$$

$$\tau_w \frac{dw(t)}{dt} = \alpha_w(V(t) - E_L) - w(t)$$

- The parameter a_w represents the level of sub-threshold adaptation
- At each spike, $w \rightarrow w + b$, which is the spike-triggered adaptation

Conductance-based AdEx (CAdEx)

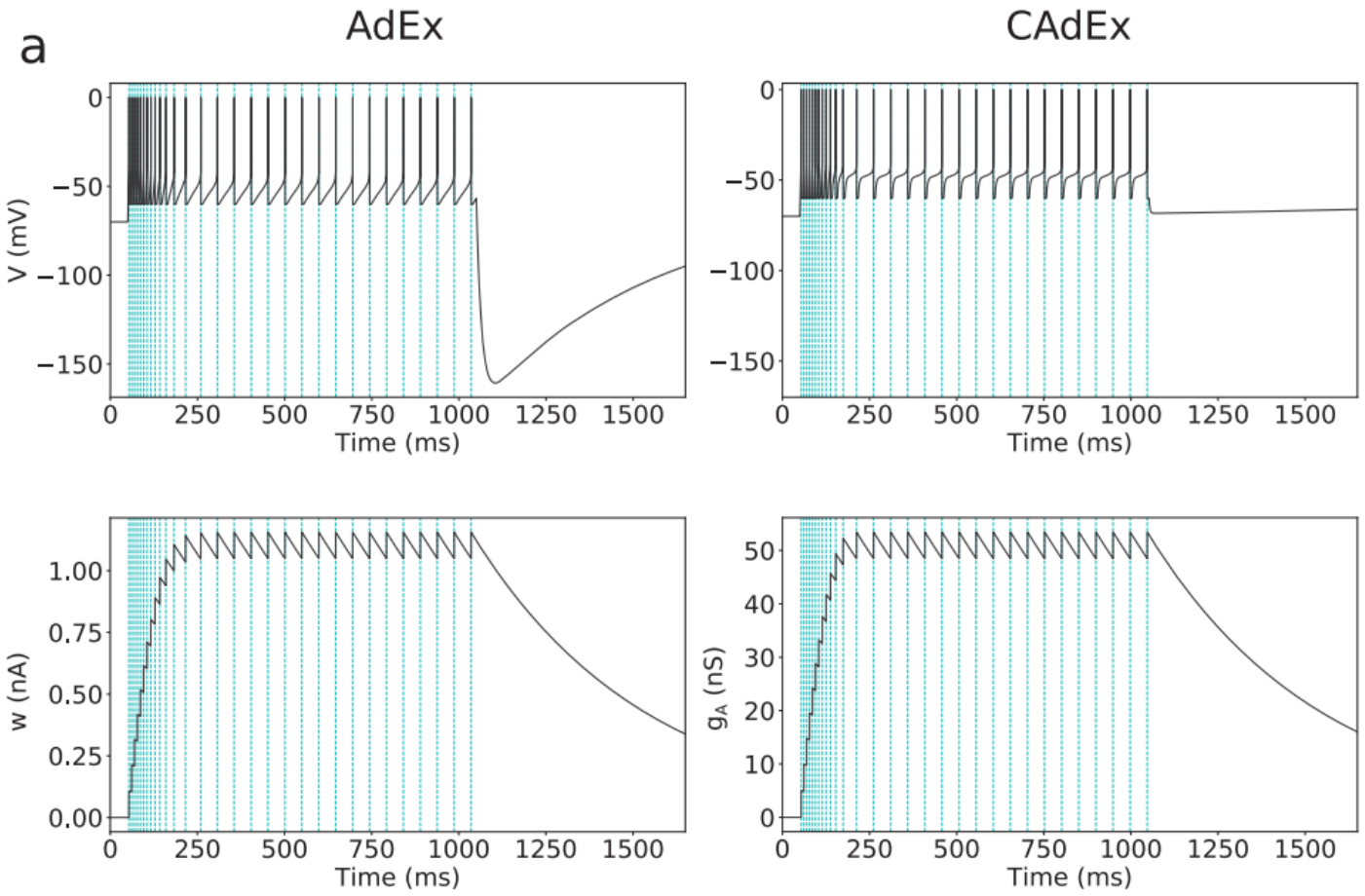
- AdEx has a low computational cost but keeping biophysical interpretation of the parameters
- However, because of its current-based adaptation, it can generate unrealistic behaviours. Górski et al., 2020 introduce CAdEx

$$C_{\mathrm{m}} \frac{\mathrm{d}V(t)}{\mathrm{d}t} = -\mathrm{g}_L(V(t) - E_L) + \mathrm{g}_L \Delta_T \exp\left(\frac{V(t) - V_T}{\Delta_T}\right) - \mathrm{g}_A(t)(V(t) - E_A) + I_e(t)$$

$$\tau_A \frac{d\mathrm{g}_A(t)}{dt} = \frac{\bar{\mathrm{g}}_A}{1 + \exp\left(\frac{V_A - V}{\Delta_A}\right)} - \mathrm{g}_A(t)$$

$$if\ V(t) \geq V_{th}: V(t) \rightarrow V_{reset}, \mathrm{g}_A(t) \rightarrow \mathrm{g}_A(t) + \delta \mathrm{g}_A$$

AdEx vs. CAdEx



Adopted from Górski et al., 2020

Izhikevich model

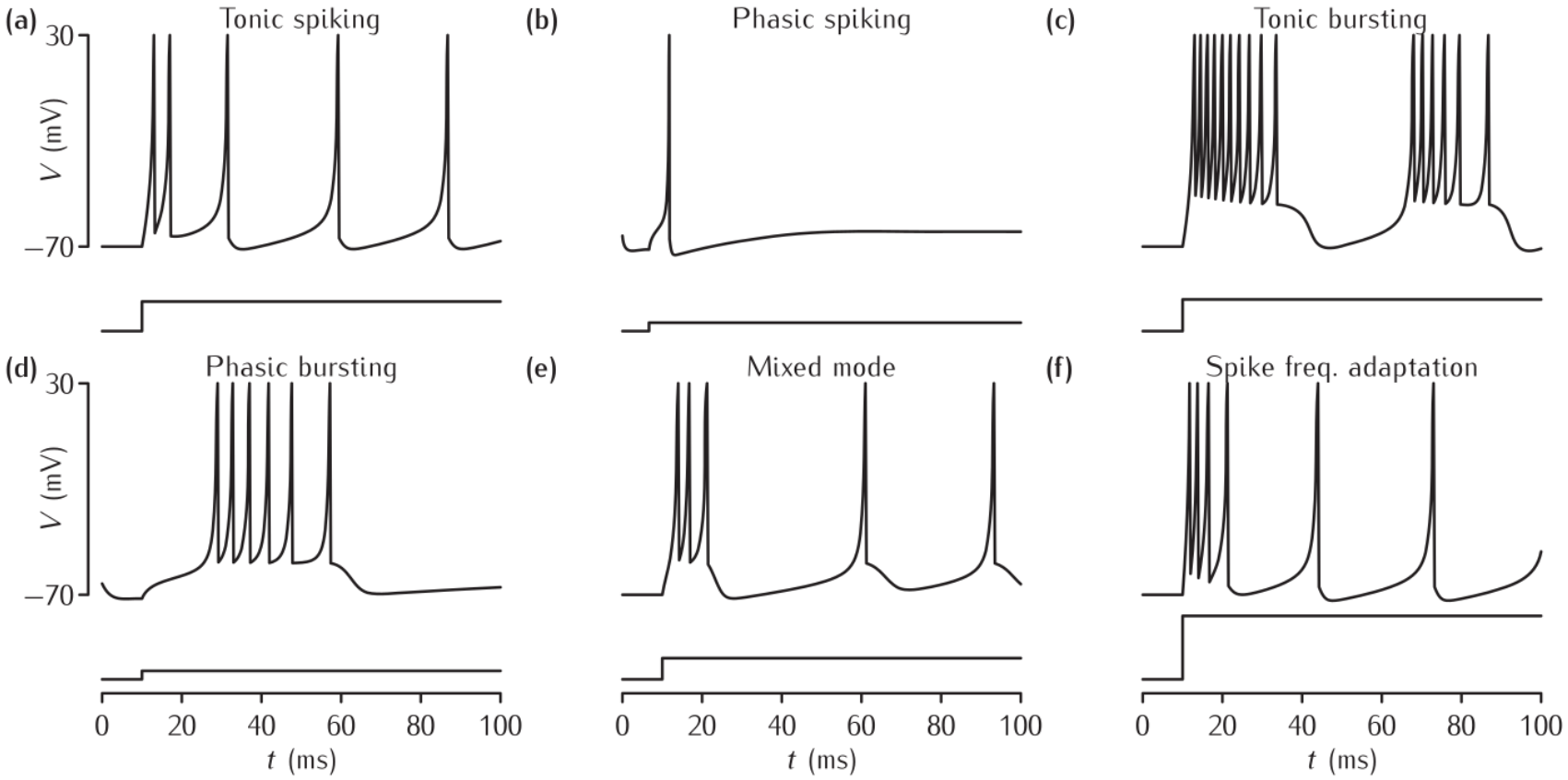
- Produces a wide range of realistic behaviors
- Added a recovery variable to the quadratic I&F

$$\frac{\mathrm{d}V}{\mathrm{d}t} = k(V - E_{\mathrm{L}})(V - V_{\mathrm{thresh}}) - u + I$$

$$\frac{\mathrm{d}u}{\mathrm{d}t} = a(b(V - E_{\mathrm{L}}) - u)$$

$$\text{if V} \geq 30\text{mV, then} \begin{cases} V \leftarrow c \\ u \leftarrow u + d \end{cases}$$

Izhikevich model



How do we fix the model parameter values?

- Step 1: Fix the known parameters (e.g. $V_{\mathrm{th}}, E_{\mathrm{L}}$, etc.) and make educated guesses for the remaining unknown parameter values.
- Step 2: Use the model to simulate experiments, producing model data.
- Step 3: Compare the model data with experimental data.
- Step 4: Adjust one or more parameter values and repeat from Step 2 until the simulated data sufficiently matches experimental data.
- Step 5: Use the model to simulate new experiments not used in the steps above and compare the resulting model data with the new experimental data to verify that the chosen parameter values are robust.

Validation (passive)

- Membrane time constant (τ_m)
- Input Resistance $\left(R_{\mathrm{IN}} = \frac{\Delta V}{\Delta I} = \frac{V_\infty - E_L}{I_e}\right)$
- Sag ratio $\left(\frac{\Delta V}{\Delta V_{min}} = \frac{V_\infty - E_L}{V_{min} - E_L}\right)$
- Rheobase current (minimum I_e to generate a spike)
- f-I curves and V-I curves